Bocconi

CORPORATE FINANCE 30017

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Introdution to the Capital Asset Pricing Model (CAPM)

Università Bocconi
MILANO

Agenda

RISK AVERSION AND PORTFOLIO CHOICE

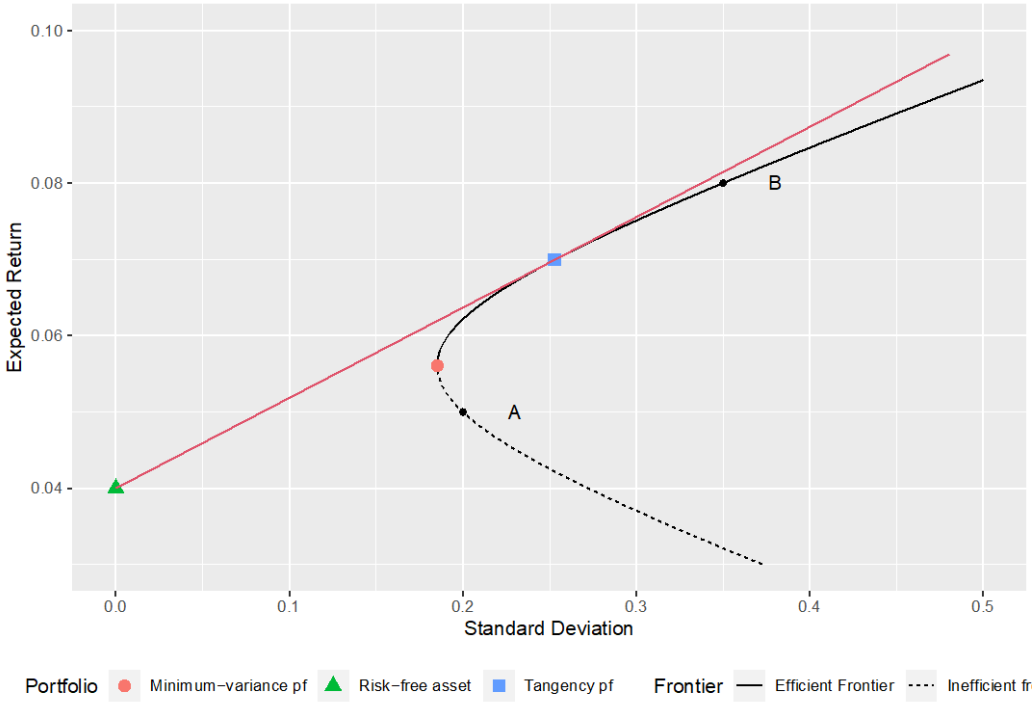
CAPM ASSUMPTIONS

CAPM PREDICTIONS

EMPIRICAL TESTS OF THE CAPM

RISK AVERSION AND PORTFOLIO CHOICE

Which Portfolio to Choose?



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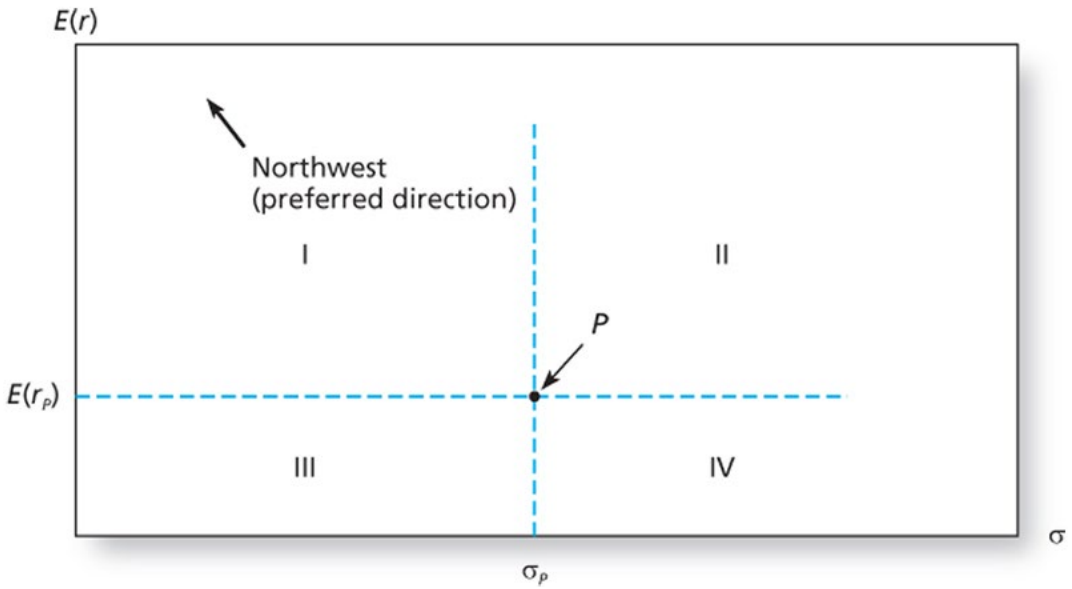
— *Along the Capital Market Line, yes…but where exactly?*

Risk Preferences

- Risk tolerance determines personal valuation discount (premium) rel. to risky payoffs expected value
  - Can estimate risk preferences based on
    - Experimental data
    - Field studies (e.g., market price differentials between similar securities with different risk)

- Assume you can buy a lottery ticket with equal chances of winning either 5 Euro or 0 Euro. What maximum price would you be willing to pay for the ticket?

Conceptualizing Preferences via Indifference Curves



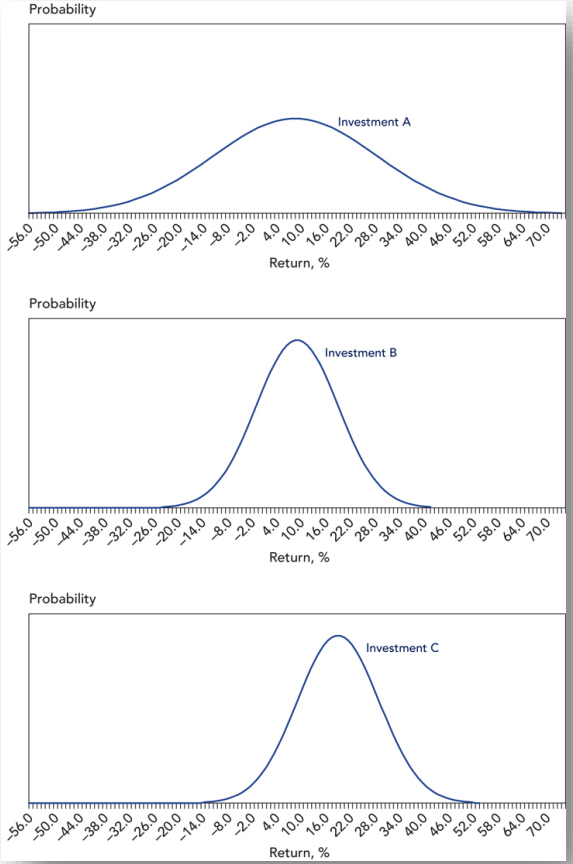
— Indifference Curve (over risk-return combinations):

- All combinations of expected return and risk that yield **same utility**

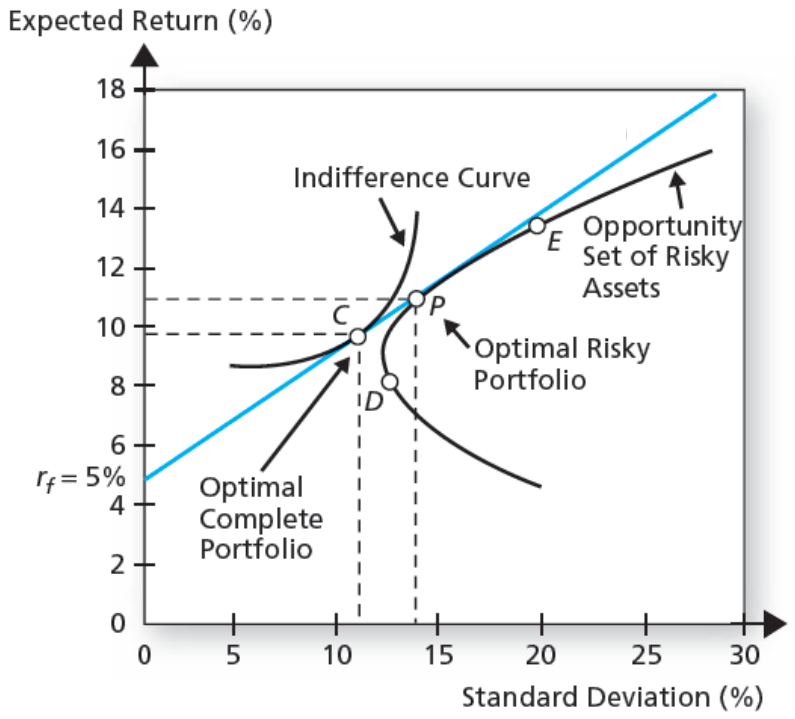
Standard assumptions

- We will assume that investors generally…
  - prefer C over B
  - prefer B over A
- Implies upward sloping indifference curve
  - Convex ind. curves: higher risk disliked disproportionately
  - Classical "Mean-Variance Preferences":

$$U(\boldsymbol{E}[r], \sigma^2) = \boldsymbol{E}[r] - \frac{\gamma}{2}\sigma^2$$



Choosing the Optimal Portfolio



- All investors choices determine prices
  - Prices pin down expected returns…
  - …in turn changing choices…
  - …
- What is the **Market Equilibrium**..?

CAPM ASSUMPTIONS

Assumptions of the CAPM

1. Individual behavior
   a. Investors are rational, mean-variance optimizers.
   b. Their common planning horizon is a single period.
   c. Investors all use identical input lists, an assumption often termed **homogeneous expectations.** Homogeneous expectations are consistent with the assumption that all relevant information is publicly available.
2. Market structure
   a. All assets are publicly held and trade on public exchanges.
   b. Investors can borrow or lend at a common risk-free rate, and they can take short positions on traded securities.
   c. No taxes.
   d. No transaction costs.

— *Which assumptions appear to be furthest off reality? Discuss.*

CAPM PREDICTIONS

Core CAPM Predictions

1. All investors **hold the same portfolio of risky assets** (the tangency portfolio).
   - Aka. "Market Portfolio" where each asset's portfolio weight equals relative market capitalization
2. Investors differ only in portfolio share of safe asset
3. Each asset priced according to

$$\boldsymbol{E[r_{i,t}] - r_{f,t} = \beta_i \times (E[r_{m,t}] - r_{f,t})}$$

where

- $r_{i,t}$ and $r_{f,t}$ denote returns during period t of risky asset i and of the risk free asset
- $r_{m,t}$ is the return during period t of the market portfolio (shorthand for $r(x_{m,t-1})$)
- β_i is the sensitivity of return of the risky asset i to the market return

Interpreting CAPM's $\boldsymbol{\beta_i}$

Excess return/risk premium
of individual stock i

$$\boxed{E[r_{i,t}] - r_{f,t}} = \boxed{\beta_i} \times \boxed{\left(E[r_{m,t}] - r_{f,t}\right)}$$

"**Quantity** of systematic risk" in stock i

"**Price** of systematic risk" (market risk premium)

Estimating CAPM's $\boldsymbol{\beta_i}$

— Regression:

$$y_t = \alpha + \beta x_t + e_t$$

where

- $y_t = (r_{i,t} - r_{f,t})$ and $x_t = (r_{m,t} - r_{f,t})$
- *Note: If CAPM holds, α should be indistinguishable from zero (statistically speaking)*

— From your statistics/econometrics courses, you know that

$$\beta_i = \frac{Cov(y_t, x_t)}{Var(x_t)} = \frac{\rho_{x,y} SD(y_t) SD(x_t)}{Var(x_t)} = \rho_{x,y} \frac{SD(y_t)}{SD(x_t)}$$

What drives $\boldsymbol{\beta_i}$ in Reality?

- Stock prices are determined by dividends (see Dividend Discount Model)
  - Hence, firms with cyclical dividends will have high betas.

- Cyclical dividends typically come from **cyclical earnings**
  - Cyclical earnings typically come from **cyclical revenues,** possibly magnified by…
    - **Operating leverage** (ratio of fixed to variable costs). Same revenue shortfall translates into larger earnings losses when fixed costs are higher.
    - **Financial leverage** (ratio of debt to equity). Since interest payments are fixed, same revenue shortfall translates into larger earnings losses when financial leverage is higher.

- Industry Betas: https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/Betas.html

The Security Market Line

$$E[r_{i,t}] - r_{f,t} = \beta_i \times (E[r_{m,t}] - r_{f,t})$$

- Within time period, different expected returns across assets must come from different β_i
  - In cross section of assets, $r_{f,t}$ and $(E[r_{m,t}] - r_{f,t})$ are fixed; only β_i varies
  - Resulting relationship between expected return and beta: Security Market Line (SML)
    - Slope of the SML is the risk premium of market portfolio
- According to CAPM, price of any asset adjusts until on SML
- *SML not to be confused with Capital Market Line:*
  - *For the CML, the x-Axis has risk, not β*

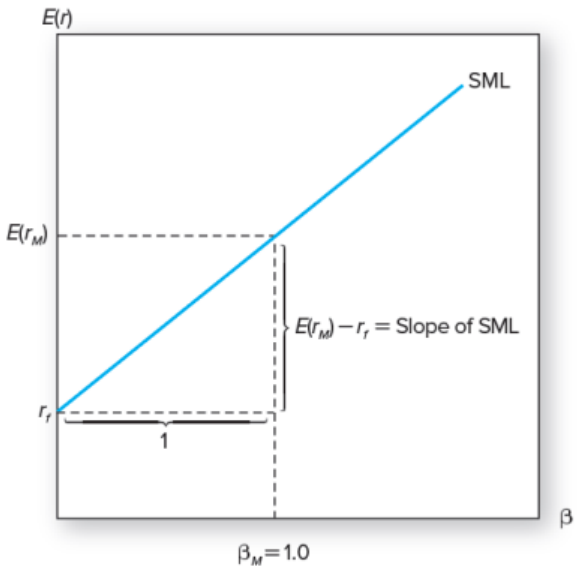
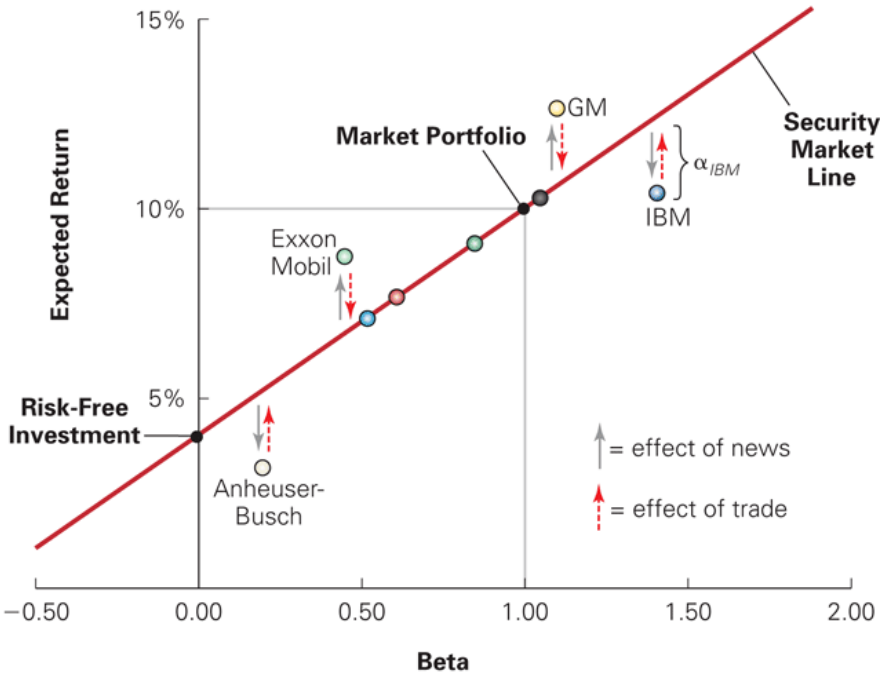


Figure 9.2 The security market line

Deviations from the SML

- *Which asset is (temporarily)…*
  - *…overpriced?*
  - *…underpriced?*
- *Why could we see deviations in practice?*



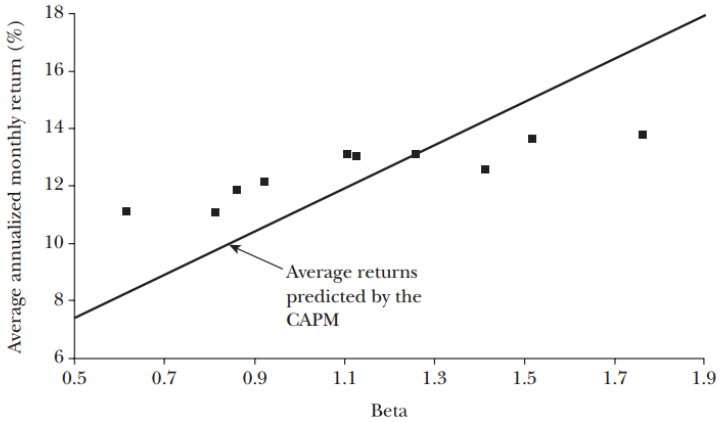
EMPIRICAL TESTS OF THE CAPM

CAPM meets Reality

- With data from the 60s and 70s, tests of the CAPM were encouraging
  - Stocks and mutual funds with higher betas produced higher returns
- Actual relationship turned out to be flatter than predicted… but still decent
  - Low-risk (high-risk) stocks appeared to earn relatively too high (low) returns

Fama and French (2004)

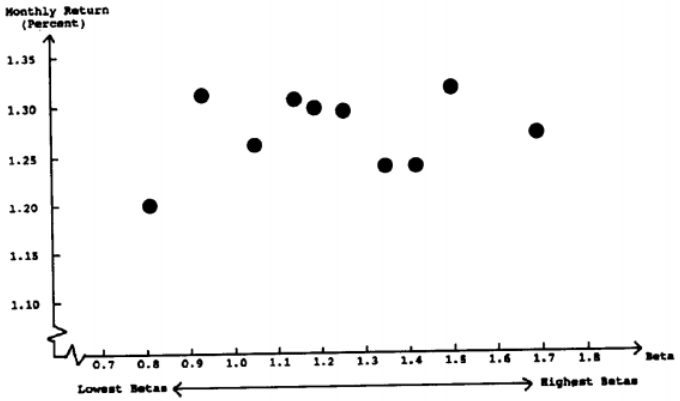
Average Annualized Monthly Return versus Beta for Value Weight Portfolios Formed on Prior Beta, 1928–2003



CAPM meets Reality, continued

- However, we can easily sort data into portfolios which will make the CAPM look a lot less impressive:
- Size: Small stocks have historically done better than big stocks
- Value: Value stocks (high b/m) have outperformed growth stocks (low b/m).
- These anomalies (and literally hundreds of others, also referred to as the “factor zoo”) cannot be justified by beta risk…

Fama and French (1992)



Fama-French Three-Factor Model

	Three-Factor Model				CAPM
	Factor Sensitivities				
	b_{market}	b_{size}	$b_{book\text{-}to\text{-}market}$	Expected Return[a]	Expected Return[b]
Autos	1.20	0.54	0.12	12.5%	11.1%
Banks	1.09	0.23	0.70	13.6	10.0
Chemicals	1.32	0.05	0.18	12.1	11.2
Computers	1.11	−0.17	−0.21	8.0	9.2
Construction	1.24	0.75	−0.07	12.5	11.6
Food	0.73	−0.33	−0.13	5.2	6.4
Oil and gas	1.03	0.17	1.10	14.9	9.6
Pharmaceuticals	1.07	0.23	−0.55	7.3	9.4
Telecoms	0.95	−0.25	0.08	8.1	8.1
Utilities	0.44	−0.22	−0.19	3.3	4.5

Concluding Remarks

- In practice, the CAPM will **never hold exactly**
  - Assumptions are violated:
    - Belief/information/expectation heterogeneity
    - Transaction cost/borrowing constraints matter
  - Moreover:
    - What exactly is the market portfolio?
    - Measurement error in beta

- Nevertheless, in industry everybody expects you to be able to apply CAPM
  - 73% of CFOs report to **always** or **almost always** use standard CAPM model, among *other* models